Chapter 9 – Software Evolution

Topics covered

✧ Evolution processes

✧ Legacy systems

✧ Software maintenance

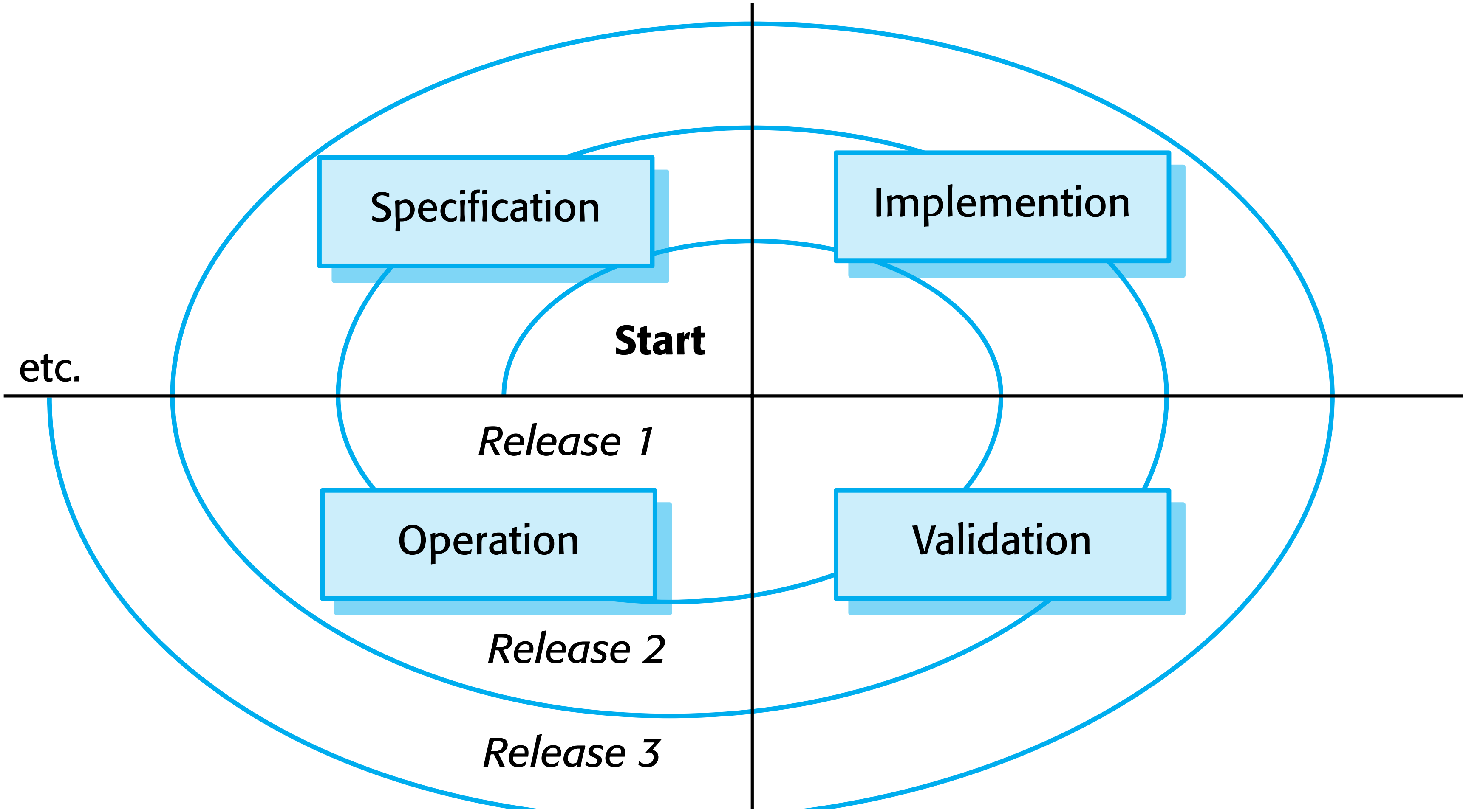
Software change

- ✧ Software change is inevitable
  - New requirements emerge when the software is used;
  - The business environment changes;
  - Errors must be repaired;
  - New computers and equipment is added to the system;
  - The performance or reliability of the system may have to be improved.
- ✧ A key problem for all organizations is implementing and managing change to their existing software systems.

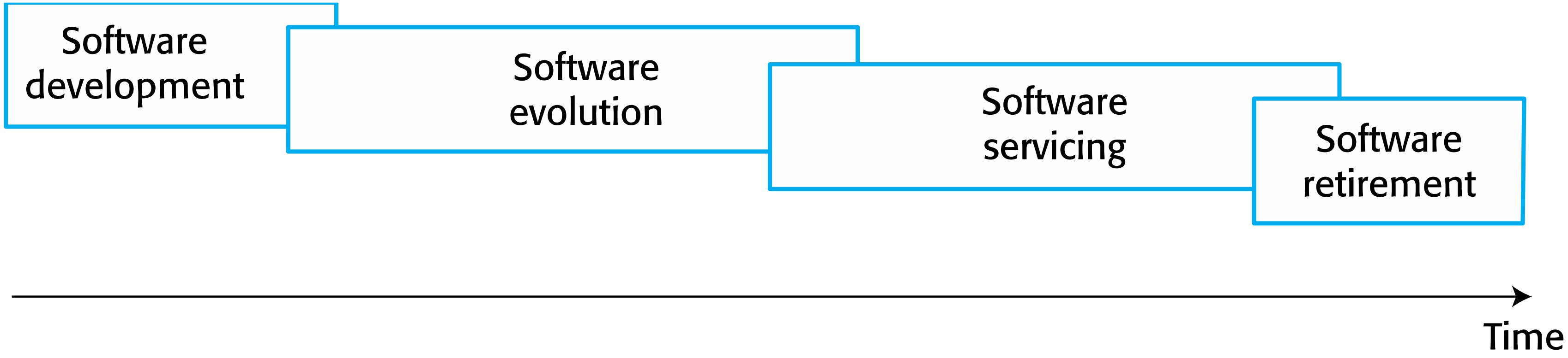
Importance of evolution

- Organisations have huge investments in their software systems - they are critical business assets.
- To maintain the value of these assets to the business, they must be changed and updated.
- The majority of the software budget in large companies is devoted to changing and evolving existing software rather than developing new software.

A spiral model of development and evolution



Evolution and servicing

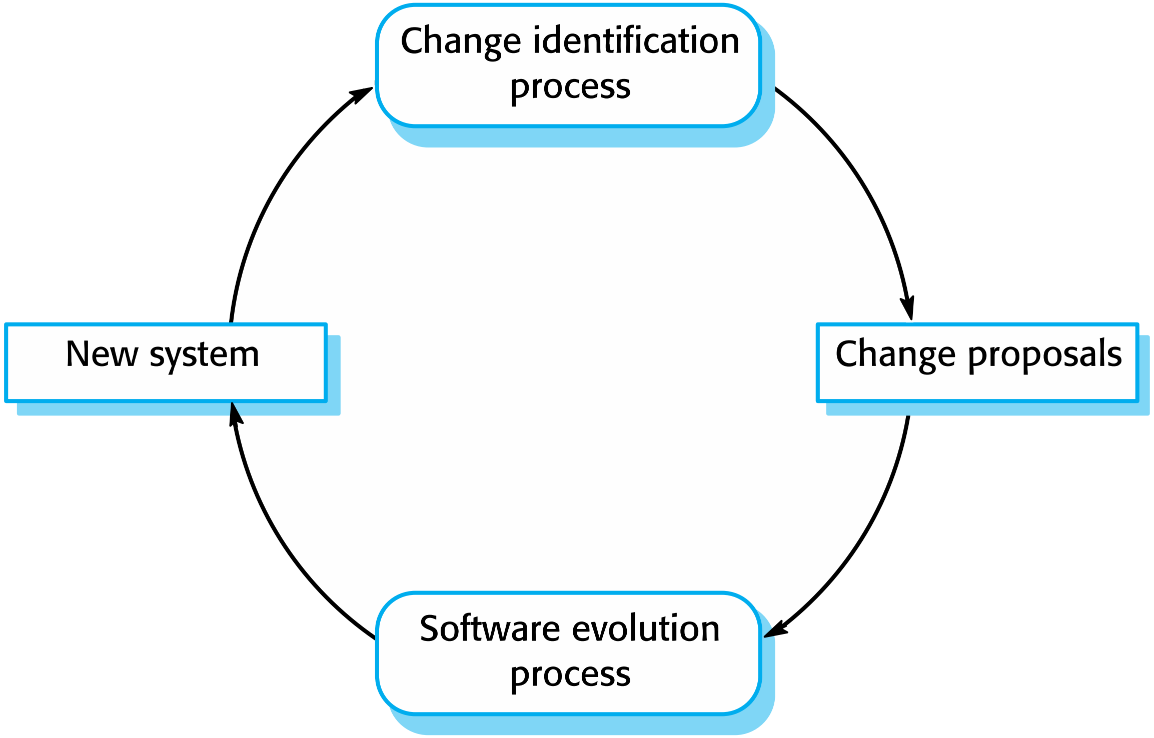


Evolution processes

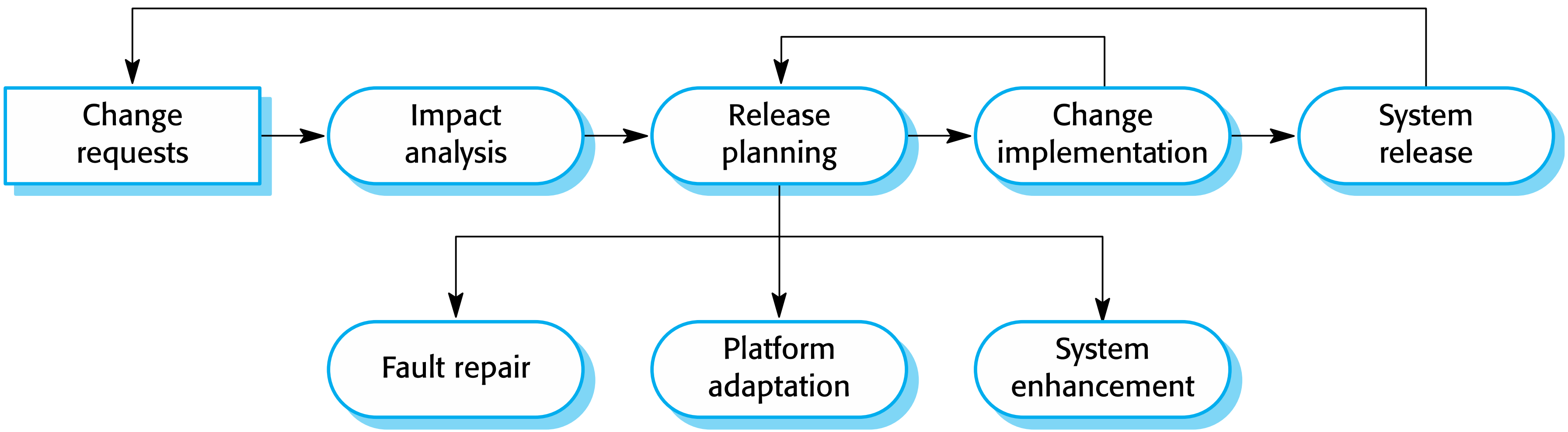
Evolution processes

- ✧ Software evolution processes depend on
  - The type of software being maintained;
  - The development processes used;
  - The skills and experience of the people involved.
- ✧ Proposals for change are the driver for system evolution.
  - Should be linked with components that are affected by the change, thus allowing the cost and impact of the change to be estimated.
- ✧ Change identification and evolution continues throughout the system lifetime.

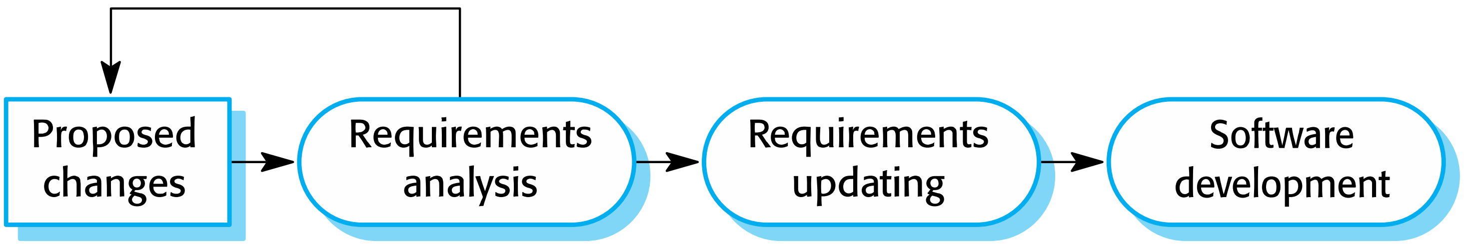
Change identification and evolution processes



The software evolution process



Change implementation



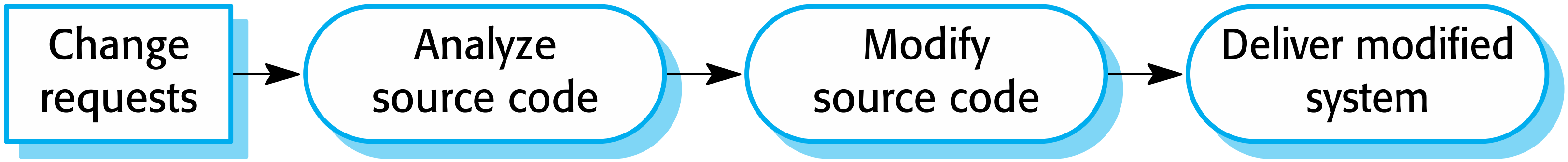
Change implementation

- Iteration of the development process where the revisions to the system are designed, implemented and tested.
- A critical difference is that the first stage of change implementation may involve program understanding, especially if the original system developers are not responsible for the change implementation.
- During the program understanding phase, you have to understand how the program is structured, how it delivers functionality and how the proposed change might affect the program.

Urgent change requests

- ✧ Urgent changes may have to be implemented without going through all stages of the software engineering process
  - If a serious system fault has to be repaired to allow normal operation to continue;
  - If changes to the system’s environment (e.g. an OS upgrade) have unexpected effects;
  - If there are business changes that require a very rapid response (e.g. the release of a competing product).

The emergency repair process



Agile methods and evolution

- Agile methods are based on incremental development so the transition from development to evolution is a seamless one.
  - Evolution is simply a continuation of the development process based on frequent system releases.
- Automated regression testing is particularly valuable when changes are made to a system.
- Changes may be expressed as additional user stories.

Handover problems

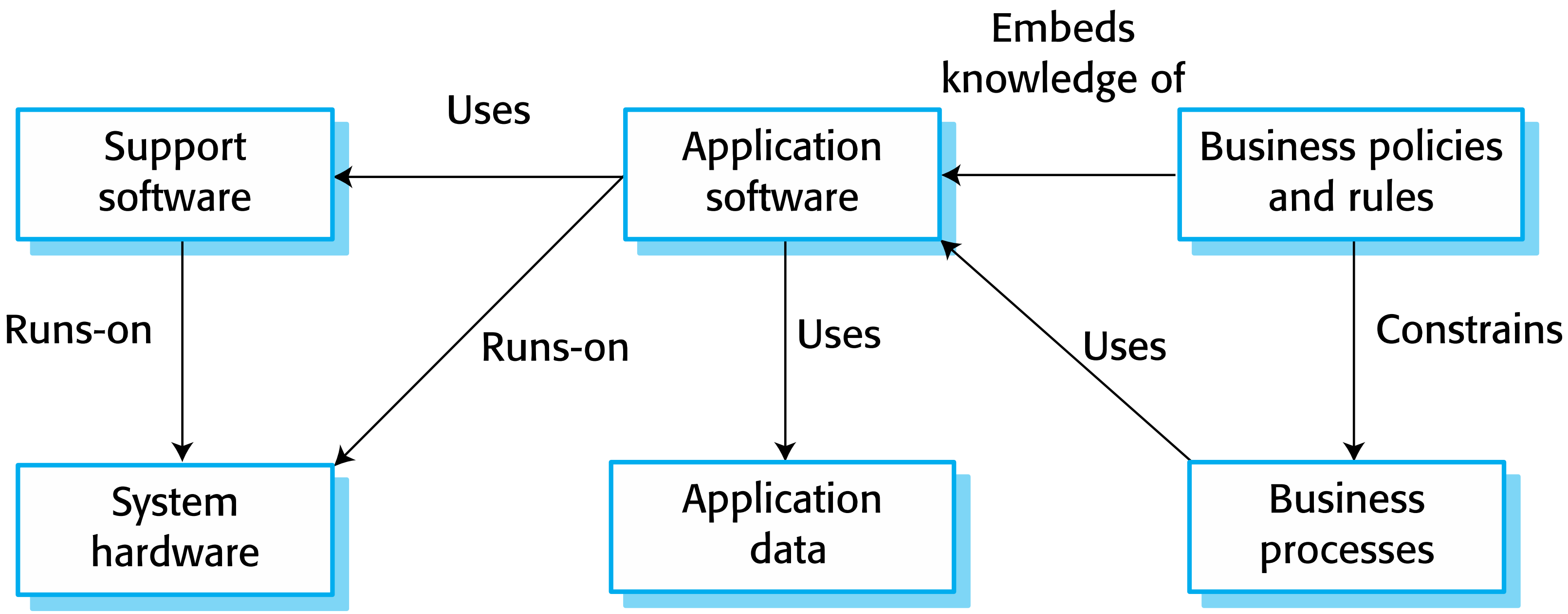
- ✧ Where the development team have used an agile approach but the evolution team is unfamiliar with agile methods and prefer a plan-based approach.
  - The evolution team may expect detailed documentation to support evolution and this is not produced in agile processes.
- ✧ Where a plan-based approach has been used for development but the evolution team prefer to use agile methods.
  - The evolution team may have to start from scratch developing automated tests and the code in the system may not have been refactored and simplified as is expected in agile development.

Legacy systems

Legacy systems

- Legacy systems are older systems that rely on languages and technology that are no longer used for new systems development.
- Legacy software may be dependent on older hardware, such as mainframe computers and may have associated legacy processes and procedures.
- Legacy systems are not just software systems but are broader socio-technical systems that include hardware, software, libraries and other supporting software and business processes.

The elements of a legacy system



Legacy system components

- *System hardware* Legacy systems may have been written for hardware that is no longer available.
- *Support software* The legacy system may rely on a range of support software, which may be obsolete or unsupported.
- *Application software* The application system that provides the business services is usually made up of a number of application programs.
- *Application data* These are data that are processed by the application system. They may be inconsistent, duplicated or held in different databases.

Legacy system components

- *Business processes* These are processes that are used in the business to achieve some business objective.
- Business processes may be designed around a legacy system and constrained by the functionality that it provides.
- *Business policies and rules* These are definitions of how the business should be carried out and constraints on the business. Use of the legacy application system may be embedded in these policies and rules.

Legacy system layers

Socio-technical system

Business processes
Application software
Platform and infrastructure software
Hardware

Legacy system replacement

- ✧ Legacy system replacement is risky and expensive so businesses continue to use these systems
- ✧ System replacement is risky for a number of reasons
  - Lack of complete system specification
  - Tight integration of system and business processes
  - Undocumented business rules embedded in the legacy system
  - New software development may be late and/or over budget

Legacy system change

✧ Legacy systems are expensive to change for a number of reasons:

- No consistent programming style
- Use of obsolete programming languages with few people available with these language skills
- Inadequate system documentation
- System structure degradation
- Program optimizations may make them hard to understand
- Data errors, duplication and inconsistency

Legacy system management

- Organisations that rely on legacy systems must choose a strategy for evolving these systems
  - Scrap the system completely and modify business processes so that it is no longer required;
  - Continue maintaining the system;
  - Transform the system by re-engineering to improve its maintainability;
  - Replace the system with a new system.
- The strategy chosen should depend on the system quality and its business value.

Software maintenance

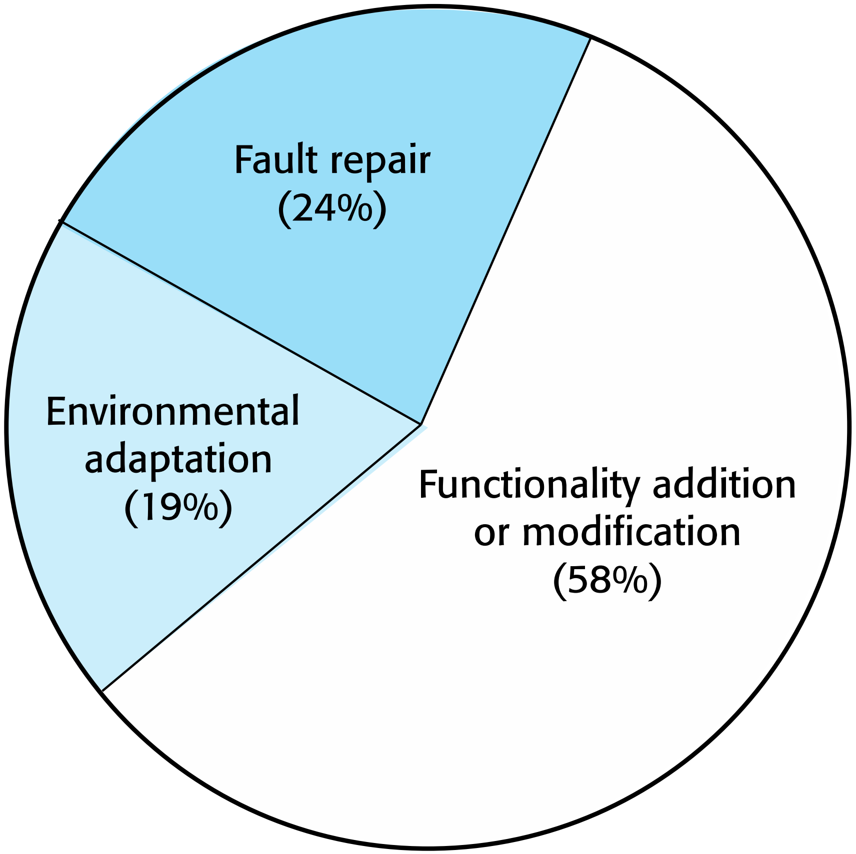
Software maintenance

- ✧ Modifying a program after it has been put into use.
- ✧ The term is mostly used for changing custom software. Generic software products are said to evolve to create new versions.
- ✧ Maintenance does not normally involve major changes to the system's architecture.
- ✧ Changes are implemented by modifying existing components and adding new components to the system.

Types of maintenance

- Fault repairs
  - Changing a system to fix bugs/vulnerabilities and correct deficiencies in the way meets its requirements.
- Environmental adaptation
  - Maintenance to adapt software to a different operating environment
  - Changing a system so that it operates in a different environment (computer, OS, etc.) from its initial implementation.
- Functionality addition and modification
  - Modifying the system to satisfy new requirements.

Maintenance effort distribution



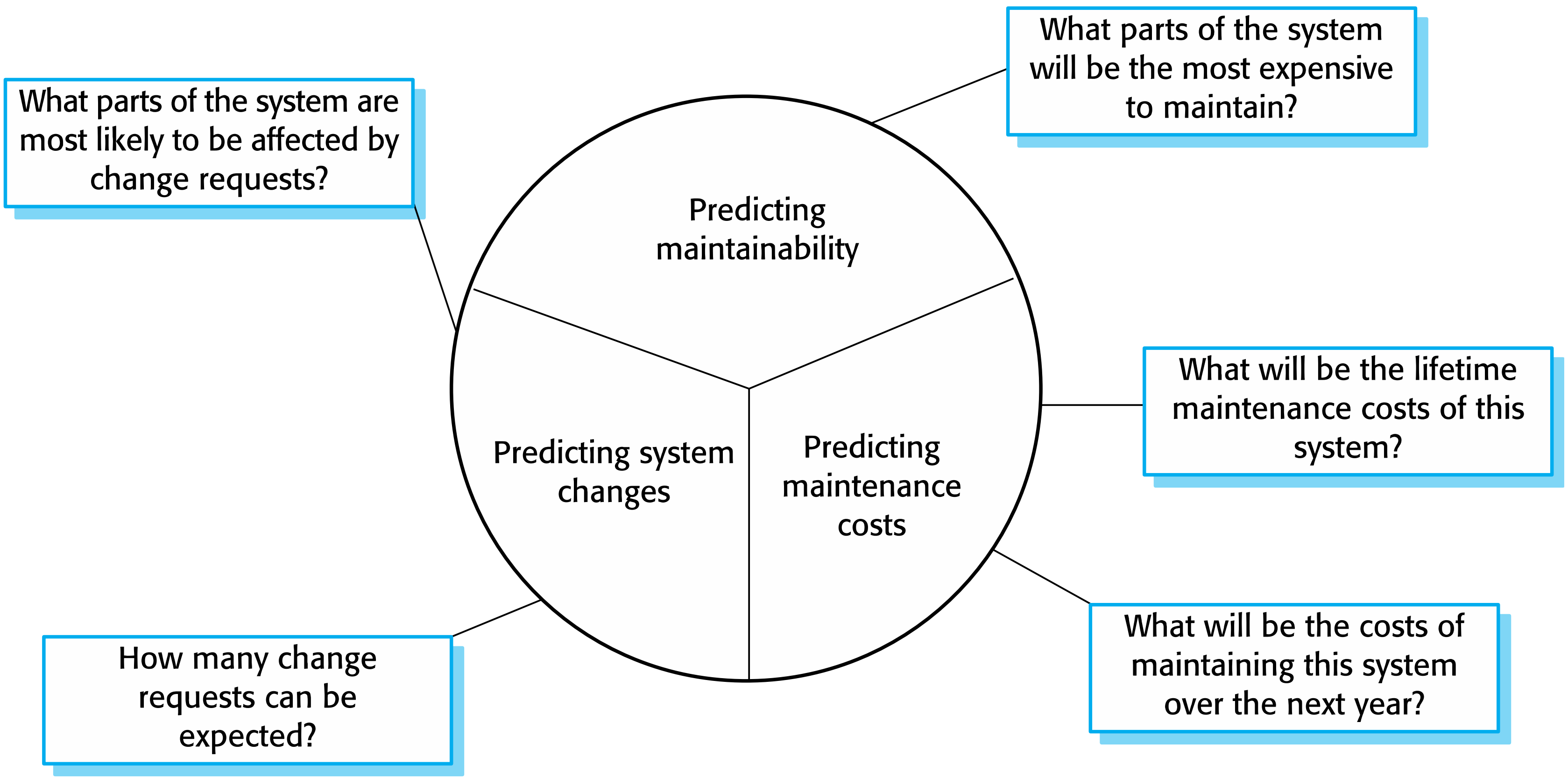
Maintenance costs

- ✧ Usually greater than development costs (2* to 100* depending on the application).
- ✧ Affected by both technical and non-technical factors.
- ✧ Increases as software is maintained. Maintenance corrupts the software structure so makes further maintenance more difficult.
- ✧ Ageing software can have high support costs (e.g. old languages, compilers etc.).

Maintenance prediction

- Maintenance prediction is concerned with assessing which parts of the system may cause problems and have high maintenance costs
  - Change acceptance depends on the maintainability of the components affected by the change;
  - Implementing changes degrades the system and reduces its maintainability;
  - Maintenance costs depend on the number of changes and costs of change depend on maintainability.

Maintenance prediction



Software reengineering

- Restructuring or rewriting part or all of a legacy system without changing its functionality.
- Applicable where some but not all sub-systems of a larger system require frequent maintenance.
- Reengineering involves adding effort to make them easier to maintain. The system may be re-structured and re-documented.

Advantages of reengineering

✧ Reduced risk

- There is a high risk in new software development. There may be development problems, staffing problems and specification problems.

✧ Reduced cost

- The cost of re-engineering is often significantly less than the costs of developing new software.

Reengineering process activities

- ✧ Source code translation
  - Convert code to a new language.
- ✧ Reverse engineering
  - Analyse the program to understand it;
- ✧ Program structure improvement
  - Restructure automatically for understandability;
- ✧ Program modularisation
  - Reorganise the program structure;
- ✧ Data reengineering
  - Clean-up and restructure system data.

Reengineering cost factors

✧ The quality of the software to be reengineered.

✧ The tool support available for reengineering.

✧ The extent of the data conversion which is required.

✧ The availability of expert staff for reengineering.

- This can be a problem with old systems based on technology that is no longer widely used.